

Harshith Goka

Research Engineer, Samsung Research Korea

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Experience

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| Samsung Research
AI System Optimization | Seoul, South Korea |
| • On Device LLM Taskforce | <i>February 2025 - Current</i> |
| Language Intelligence Team | <i>February 2024 - January 2025</i> |
| Visual Intelligence Team | <i>March 2023 - February 2024</i> |
| | <i>September 2019 - March 2023</i> |
| Samsung Research | Seoul, South Korea |
| • Internship @ Vision Team | <i>Summer 2018</i> |

Education

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| Indian Institute of Technology, Bombay | Mumbai, India |
| • B.Tech. in Computer Science and Engineering; CGPA: 8.74/10 | <i>July 2015 - June 2019</i> |
| – Major: Computer Science and Engineering | |
| – Key Electives: Foundations of Intelligent and Learning Agents, Advanced Machine Learning, Foundations of Machine Learning, Computer Vision, Design and Re-engineering of Object Oriented Programming, Parallel Programming Paradigms, Foundations of Digital Image Processing | |
| – Core Courses: Artificial Intelligence, Implementation of Programming Languages, Automata Theory, Discrete Structures, Data Analysis and Interpretation, Abstractions and Paradigms for Programming, Databases and Information Systems, Software Systems Lab, Operating Systems, Data Structures and Algorithms, Design and Analysis of Algorithms, Computer Networks, Digital Logic Design, Computer Architecture | |

Skills

- **Languages:** Python, C/C++, Java, Kotlin, $\text{\LaTeX}$, Dart, JavaScript, SQL, VHDL, C#
- **Frameworks:** Android SDK, MATLAB, Docker, Flutter, Qualcomm AI Direct Engine(QNN)
- Experienced in using popular deep learning libraries **PyTorch**, **JAX**, **Tensorflow**, **Transformers** on multinode HPC clusters

Key Projects

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| On-Device LLM Neural Architecture Search | Samsung Research, Korea |
| • Leader: Lee Kyenghun | <i>June 2025 - Current</i> |
| – Developing a framework for automatic configuration of architectures for pretraining LLM models using NeMo. | |
| – Conducting various heuristic based model structure trainings for improving the quality while reducing latency or model size. | |
| – Also using various search algorithms to automatically improve upon existing architectures for on device LLM architectures which are both fast and high quality. | |

- **Optimization of On-Device Architectures** Samsung Research, Korea
Feb 2025 - Current
 - Leader: Lee Kyenghun
 - Developed a framework for NPU latency testing.
 - Optimized various architectures like state space, hybrid and new prototype architectures for on-device inference.
 - Also worked on optimizing a speech encoder for LLM and TTS diffusion models for the NPU.
- **Quantization for On Device LLMs** Samsung Research, Korea
March 2024 - July 2024
 - Leader: Lee Kyenghun
 - Mainly worked on quantization of LoRA with Samsung Gauss On-Device models.
 - Improved various tools used for quantization of deep models.
- **Alleviating Hallucination in News Summarization Models** Samsung Research, Korea
May 2023 - January 2024
 - Leader: Ko Hyeonmok
 - Worked on reducing the occurrence of hallucinations in News Summarization models.
 - Experimented on using various scoring functions for aligning the model.
- **Object Eraser - Video Inpainting** Samsung Research, Korea
February 2022 - December 2022
 - Leader: Naejin Kong
 - Explored video inpainting models using Transformer in the Generative Adversarial Network (**GAN**) framework.
 - Experimented on various **Vision Transformers** model types to improve the quality.
- **Object Eraser - Image Inpainting** Samsung Research, Korea
March 2021 - December 2021
 - Leader: Naejin Kong
 - Explored various ways to use Generative Adversarial Networks (**GANs**) to inpaint part of image to match the background.
 - Co-developed a new way of using Fast Fourier Convolutions to improve the quality of large mask inpainting.
 - Developed various sized on-device models to run on mobile phones and laptops using **TFLite** and **ONNXRuntime**.
- **Image Relighting** Samsung Research, Korea
September 2019 - March 2020
 - Leader: Naejin Kong
 - Explored various ways of relighting an image by detecting types of materials and re-rendering them.
 - Generated a large amount of synthetic data for detecting materials.
 - Trained deep U-net models to extract various component maps like albedo, normal and roughness maps.
- **Attention Methods for Deep Metric Learning** Samsung Research, Korea
Summer 2018
 - Guide: Wonsik Kim
 - Explored different soft attention methods for deep metric learning on images.
 - Developed GPU code for a few attentions models using **Caffe**.
 - Evaluated these different models on the dataset of clothes using the **Recall@K** metric.

Scholastic Achievements

- Secured **All India Rank 25** in **JEE (Advanced)** out of **150,000** candidates (2015)
- Secured **All India Rank 8** in **JEE Main (B.Arch)** out of **150,000** candidates (2015)
- Secured **All India Rank 26** in **JEE Main (B.Tech)** out of **1.3 million** candidates (2015)
- Awarded the Kishore Vaigyanik Protsahan Yojana (**KVPY**) Fellowship held by the Department of Science and Technology, Government of India with an **All India Rank 70** (2014)
- Secured **State Rank 19** in **AP State Common Entrance Test** out of about **150,000** candidates (2015)
- Secured **State Rank 60** in **Telangana State Common Entrance Test** out of about **250,000** candidates (2015)
- Awarded **AP grade** for excellence (given to top 1% students) in Engineering Drawing (2015-16)